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1 September 1969
Superseding
MIL-I-22062(AER)
15 June 1959

MILITARY SPECIFICATION

INTERCONNECTING BOX J-1013A/AIC

This Specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers Interconnecting Box J-1013A/AIC for use in Military equipment.

1.2 Classification - The Interconnecting Box J-1013A/AIC shall be of one type and shall consist of the following:

Type Designation	Applicable Paragraph
Interconnecting Box J-1013A/AIC	3.3.1 3.4.3

1.3 Associated Equipment - The J-1013A/AIC Interconnecting Box shall operate with the Associated Equipment listed in 6.6.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-14	Molded Plastics and Molded Plastic Parts, Thermosetting
MIL-C-172	Cases;Bases, Mounting; and Mounts Vibration (For use with Electronic Equipment in Aircraft)

FSC 5831

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MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definitions and Systems of Units, Electromagnetic Interference Technology
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for System and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND

NAVWEPS 16-1-525 Avionics Preferred Standard Test Equipment

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2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

3. REQUIREMENTS

3.1 Qualification - The interconnecting box furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Parts and Materials

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.3 Design and Construction - The interconnecting box shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the interconnecting box, excluding cables, shall be a minimum consistent with good design and shall not exceed 12 ounces.

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The interconnecting box shall operate with satisfactory performance, continuously or intermittently for a period of at least 1000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

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3.3.2.2 Operating Life - The interconnecting box shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The interconnecting box shall have 500 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of paragraph 4.7.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The interconnecting box shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The interconnecting box shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the interconnecting box.

3.3.4 Interchangeability - For reordered interconnecting boxes interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-STD-461, 462, 463.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used as a basis to establish normal performance charateristics under standard conditions and for making laboratory bench tests.

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Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity

3.3.9 Service Conditions - The interconnecting box shall operate satisfactorily under any of the environmental service conditions or reasonable combinations of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment.

3.3.9.1 Vibration - The interconnecting box shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.3.10 Insulating and Case Materials - Insulating and case materials shall comply with the following requirements, as applicable.

3.3.10.1 Case - The case shall be of aluminum alloy and shall conform to the applicable requirements of Specification MIL-C-172. The cover shall be removable.

3.3.10.2 Potting - The components shall be encapsulated within the case of the interconnecting box with the exception of the movable link switches mounted on the terminal board. The switches shall be readily accessible for positioning.

3.3.10.2.1 Plastic-Material - The plastic-material used for potting shall conform to the requirements of Specification MIL-M-14. Type MFE shall be used where electrical properties are predominant.

3.3.10.3 Insulating and Impregnating Compounds - All such compounds including varnishes, waxes, and the like, shall be suitable in every respect for each particular application. Under all service conditions including test conditions outlined in Section 4 of this specification, the compound shall adequately preserve the dielectric strength of the insulation to which it is applied. A compound shall not, either in the state of its original application or as a result of aging under severe service conditions, show any injurious effect upon the insulation it is designed to protect, and shall not cause corrosion or deterioration of adjacent metal or plastic parts.

3.3.11 Cleaning and Finishing of Metal Parts -

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3.3.11.1 Cleaning - Where cleaning operations on metal parts are not specified in detail, they shall be in accordance with the best suitable commercial practice.

3.3.11.2 Aluminum Alloys - Where aluminum alloy parts are employed they shall be cleaned by immersion in a hot caustic or other suitable solution until etched, rinsed thoroughly in clean cold water, immersed in nitric acid until the remaining caustic is neutralized, and thoroughly rinsed in cold, then hot water. Anodic treatment is not prohibited, but masking shall be employed where necessary to retain the quality of the original metal at areas where good electrical contact is necessary.

3.3.11.3 Other Non-Ferrous Metals and Alloys - Copper, Brass, Bronze, Berryllium-Copper and other similar materials shall be plated or otherwise suitably finished or enclosed to provide protection against corrosion and deterioration. The kind of plating or finish applied shall depend upon the characteristics of the individual part and its use in the interconnecting box. The plating or finish used for each individual part shall be such as to provide good appearance, corrosion and deterioration resistance, and long life for that part. Various problems such as soldering, fitting, assembly, reaction with adjacent parts and the like, must be considered in the selection of the plating or finish.

3.3.12 Input Power Requirements - The input power for the amplifier of the J-1013A/AIC shall be derived from the equipment with which the interconnecting box is associated (i.e., intercommunicating equipment or transmitting equipment). When operating with such equipment the output terminals of the interconnecting box serve also for the D. C. power supply input. The preamplifier shall be polarity sensitive. For test purposes the interconnecting box shall be connected as shown in Figure 4.

3.4 Detail Requirements - The design and assembly of the interconnecting box shall be such that the unit is inherently stable as to mechanical and electrical characteristics. The detailed mechanical and electrical design of the interconnecting box shall be accomplished by the contractor, subject to the requirements of this specification. The requirements are detailed herein only to the extent considered necessary to obtain the desired mechanical and electrical characteristics and performance, and permanence of the same.

3.4.1 Function - The interconnecting box shall permit the intercommunication set and/or radio in an aircraft to be operated with an optional combination of headsets and microphones. This function shall include acceptance and delivery of headset signals, monaural or split phone application, and/or of microphone signals without change in either signal or impedance levels. This function shall also include transformation of high impedance level headset signals to low impedance level for either monaural or split phone applications. Finally, this function shall include amplification of low level dynamic microphones and delivery of such amplified microphone signals at an impedance level of 100 ohms.

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3.4.1.1 Operation with Accessories - When the movable link switches of the interconnecting box are positioned and fastened in accordance with the operating chart within the cover as shown in Figure 2, the circuits shall permit the use of the extension cord assemblies in 6.6 with the possible accessory equipments described in Figure 3.

3.4.2 Contents - The Interconnecting Box J-1013A/AIC, shall contain the following components shown in Figure 2:

3.4.2.1 Link Switches - Switches of the sliding link type shall be mounted on the terminal board of the unit. Each switch shall be positioned and locked by nickel plated brass screws. The position of the switches shall determine the circuits usable within the unit.

3.4.2.2 Audio Transformers - Two audio transformers, isolating impedance matching type, shall provide a nominal impedance in the primary windings of 300 ohms each. The secondary windings shall have a nominal impedance of 15 ohms. Various headsets are usable from the output of the intercommunication set and/or radio when the switches are positioned in order to permit the transformers to be used.

3.4.2.3 Preamplifier - A preamplifier assembly shall convert the *output of a 5 ohm dynamic-type microphone to the power required from a 60-100 ohm* carbon type microphone when the switches are positioned to permit the use of the preamplifier. The preamplifier shall be powered by the microphone current supplied from associated utilization equipment. When the dynamic-type microphone is used in connection with the interconnecting box, additional preamplification is not required.

3.4.2.4 Connectors - The pygmy-type connectors J1 and J2 (Bendix or approved equivalent) shown in Figure 1, shall be used to provide all external connections to this unit. Input and output connections shall be as indicated in Tables A and B respectively of Figure 1.

3.4.3 Form Factor and Mounting - The interconnecting box shall have the dimensions, construction features and provisions to mount the unit to the airframe as described in Figure 1.

3.4.4 Failure Caused by Switching Action - In the event that any switch is operated simultaneously with any other switch or switches, no failures shall be caused thereby.

3.4.5 Instructions - An operating chart shall appear on the inner side of the cover in the form of a decalcomania, or similar means.

3.4.6 Distortion - The distortion for the preamplifier of the microphone circuit and the audio transformers of the headset circuits shall be as specified in 3.4.9.6 and 3.4.10.3 respectively.

3.4.7 Overload - Each headset circuit shall withstand the overload test in 4.5.2.3. This test shall first be performed on the circuits which include the transformers and then on the circuits with the transformers bypassed.

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3.4.8 Direct Current Resistance - The direct current resistance of any line shall not exceed 0.5 ohm when measured from the required pins of the connectors. The transformers and preamplifier shall be bypassed by the switching procedure described in Figure 2 prior to the tests.

3.4.9 Preamplifier of the Microphone Circuit - The preamplifier, included in the microphone circuit of Figure 2, shall meet the following requirements:

3.4.9.1 Gain - Under standard conditions, the preamplifier shall have a voltage gain of 61 ±1 db at 1000 cps. When the preamplifier is operated subject to any combination of service conditions, the output shall not vary more than ±3 db, referenced to the output under standard conditions, except when operated between -35°C, and -54°C, the output may be down a maximum of 6 db.

3.4.9.1.1 Gain Stabilization Time - The preamplifier gain of 61 ± 1 db at 1000 cps shall be attained within 1 db in 100 milliseconds or less after the application of DC input power.

3.4.9.2 Impedance - The input impedance of the preamplifier shall be 5 ±1 ohms at 1000 cps. The output impedance shall be 100 ±20 ohms at 1000 cps under standard or service conditions when the input utilizes a 5 ±1 ohms load.

3.4.9.3 Signal Input Voltage - The nominal signal input voltage shall be 250 microvolts ±5%. This input voltage shall be used for test purposes unless otherwise specified.

3.4.9.4 Linearity - The output voltage of the preamplifier, when properly terminated, shall be a linear function, within ±10%, when the input voltage is varied between 10 and 500 microvolts.

3.4.9.5 Frequency Response - The frequency response shall be 61 ±1 db between 400 and 2250 cps, with the signal input voltage as specified in 3.4.9.3. The response at 400 cps and below and at 2250 cps and above shall be in accordance with Table I.

TABLE I

Frequency (cps)	Minimum Gain (db)	Maximum Gain (db)
50	--	52
100	--	56
300	59	62
400	60	62
Thru 2250	60	62
2500	59	62
6000	--	56
10,000	--	50
25,000	--	38
50,000	--	26
100,000	--	14

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3.4.9.6 Distortion - The distortion shall not exceed 2.0% between 300 and 2500 cps with the standard input voltage as specified in 3.4.9.3.

3.4.9.7 Amplifier Stability - The preamplifier shall be stable and shall show no evidence of regeneration or oscillation when operated with input and output circuits terminated in the required impedance. No damage shall result from preamplifier operation with input and/or output terminated in either open or short circuit. The preamplifier shall meet the gain characteristics of 3.4.9.1 when the voltage of the power source is varied between the limits of 22 and 30 volts.

3.4.9.8 Overload Protection - The preamplifier shall not be damaged by overvoltage resulting from short circuit of the 490 ohm ±5% resistance of the power source, (Figure 4). The preamplifier shall resume normal operation when the voltage is returned to operating limits.

3.4.9.9 Under Voltage Protection - The preamplifier shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits specified.

3.4.9.10 Voltage Output - The preamplifier shall be capable of providing a minimum of 0.9 volt rms with less than 10% peak distortion.

3.4.9.11 Internal Noise - Noise level referred to the input shall not exceed 1 microvolt when tested in accordance with 4.5.2.4.11.

3.4.9.12 Input Current - The preamplifier shall require less than 14 ma when operated in accordance with Figure 4.

3.4.9.13 Susceptibility

3.4.9.13.1 Conducted R.F. Susceptibility - The preamplifier shall meet the requirements of Table II when tested using the circuit shown in Figure 4. The input signal shall be 30,000 microvolts modulated 40% at 400 cps.

TABLE II

Test Frequency	Maximum Allowable Output
0.10 MC	5.5 millivolts
0.50 MC through 10.0 MC	1.5 millivolts

3.4.9.13.2 Radiated R.F. Susceptibility - The preamplifier, when connected as shown in Figure 4, shall not evidence oscillation or other degradation of performance including extraneous signals or noise (i.e. output which does not correspond to a desired input) when subjected to any frequency and steady field intensity shown in Table II A. The preamplifier shall not evidence oscillation or other degradation when subjected to the required steady field intensity for a period of three minutes at nine frequencies including end and mid frequency of each frequency band of Table IIA.

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TABLE II A

Frequency (MC)	Steady Field Intensity (Antenna Induced)
2-30	-4db relative to 1.0 volt (0.63 volt) in a 41 inch vertical rod antenna*
30-52	-2db relative to 1.0 volt (0.8 volt) in a half wave dipole antenna
225-400	24 db relative to 1.0 volt (16.0 volt) in a half wave dipole antenna

* As used with NF-105 Noise & Field strength Meter or Equivalent

3.4.9.14 Magnetic Susceptibility - The interconnecting box shall be shielded so that when oriented in any position within a magnetic field of 2 gauss/cm^2 in the frequency range of 400 to 2800 cps, the output from the preamplifier shall not be greater than 5 millivolts at 400 cps or 50 millivolts at 2800 cps. Between 400 cps and 2800 cps the output shall not exceed the limits established by a linear curve connecting the two extremes specified above.

3.4.9.15 Ripple Susceptibility - The preamplifier, when connected in the circuit shown in Figure 5, shall have an output level equal to or less than the test signal. The test frequency shall be 100 cps to 10,000 cps and the signal level shall be 100 millivolts.

3.4.9.16 Voltage Transients Susceptibility - There shall be no degradation of performance or any change in DC current of the preamplifier as a result of the Voltage Transients Test in accordance with paragraph 4.5.2.4.18.

3.4.9.17 Operation - The preamplifier shall provide satisfactory operation and shall not adversely affect the performance of any other component of Interconnecting Box J-1013A/AIC when connected with and operated as part of that equipment.

3.4.10 Audio Transformers of the Headset Circuits - The audio transformers of the headset circuits shown in Figure 2 shall meet the following requirements:

3.4.10.1 Power Data - The transformers shall each provide a power transfer of at least 500 milliwatts with a power insertion loss not to exceed 2.5 db.

3.4.10.2 Frequency Response - The frequency response characteristics of each transformer circuit shall be smooth and free from abnormal irregularities over the frequency range of 300 to 7000 cps. The output response over this frequency range shall be within $\pm$ 3 db of the 1000 cps output.

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3.4.10.3 Distortion - The distortion introduced by the transformers shall not exceed 5 percent each for frequencies of 1000 to 7000 cps. The distortion may increase for frequencies below 1000 cps but shall not exceed 10 percent in the range of 1000 to 300 cps.

3.4.10.4 Impedance - The impedance of each primary transformer winding shall be 300 ohms ±10% and the impedance of each secondary winding shall be 15 ohms ±10% when measured at 1000 cps. The impedance of the primary and secondary windings over the frequency range of 300 to 7000 cps shall not deviate from the 1000 cps values by more than 10% when properly terminated.

3.4.10.5 Operation - The transformers shall provide satisfactory operation and shall not adversely affect the performance of any other component of Interconnecting Box J-1013A/AIC when interconnected with and operated as part of that equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - The interconnecting box shall be subjected to the following tests to determine compliance with applicable requirements.

(1) Qualification Tests - Qualification Tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(2) Acceptance Tests - Acceptance Tests are those tests accomplished on interconnecting boxes manufactured and submitted for acceptance under the contract.

(a) Individual Tests - Individual Tests are those tests conducted on each interconnecting box submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of interconnecting boxes as approved by the U.S. Naval Ammunition Depot, Crane, Indiana, and which have previously passed Individual Tests.

(3) Life Tests - Life Tests shall be conducted by the contractor at his plant. These tests shall be started as soon as practicable on an early production interconnecting box.

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4.2 Qualification Tests - Qualification Tests are to be made on samples submitted for qualification as satisfactory production units. These tests shall be conducted at a laboratory and under conditions satisfactory to the U.S. Naval Ammunition Depot and shall consist of the tests specified in Table III. Ten units shall be submitted for qualification tests. Modifications required as a result of these tests shall be incorporated in the amplifier prior to resubmission for qualification requests for authorization of qualification tests shall be addressed to the U.S. Naval Ammunition Depot (See 6.5).

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for all tests performed under the supervision of the cognizant Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government for sampling tests shall engage the services of a commercial testing laboratory acceptable to the U.S. Naval Ammunition Depot. The contractor shall furnish test reports, in duplicate, via the cognizant Government Inspector, showing quantitative results for all tests required by this specification and signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance of the finished product shall be upon a lot basis after satisfactory completion of all required tests. Each lot shall be composed of interconnecting boxes which have successfully passed the individual tests. Acceptance tests shall consist of the following:

4.3.1 Individual Tests - Each interconnecting box submitted for acceptance shall be subjected to the individual tests to determine compliance with requirements of material, workmanship and operational adequacy. As a minimum each interconnecting box shall be subjected to the tests specified in Table III.

4.3.2 Sampling Tests - Sampling Tests shall be conducted on samples from each lot as specified in the following paragraph.

4.3.2.1 Sampling Instructions - One interconnecting box from the first 10 production interconnecting boxes, and thereafter two from each lot of 100, or fraction thereof, as selected by the cognizant Government Inspector, shall be given the specified sampling tests of Table III.

4.4 Tests Required - The interconnecting box shall be subjected to the tests specified in Table III and any other tests deemed necessary by the U.S. Naval Ammunition Depot or the cognizant Government Inspector to determine compliance with the requirements of this specification.

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TABLE III

Tests	Test Procedure Para.	Reqmt. Para.	Qual. Tests	Ind. Tests	Sampl. Tests
Examination of Product	4.5.2.1	3.3.3	x	x	x
		3.4.3	x	x	x
		3.3.10	x	x	x
Operational Test	4.5.2.2	3.4.9.17	x	x	x
		3.4.10.5	x	x	x
Overload Measurement	4.5.2.3	3.4.7	x		x
		3.4.10.1	x		x
Gain	4.5.2.4.1	3.4.9.1	x	x	x
Gain Stabilization	4.5.2.4.2	3.4.9.1.1	x		x
Impedance	4.5.2.4.3	3.4.9.2	x		x
	4.5.2.5.3	3.4.10.4	x		x
Linearity	4.5.2.4.4	3.4.9.4	x		x
Frequency Response	4.5.2.4.5	3.4.9.5	x	x	x
	4.5.2.5.1	3.4.10.2	x	x	x
Distortion	4.5.2.4.6	3.4.9.6	x	x	x
	4.5.2.5.2	3.4.10.3	x	x	x
Amp Stability	4.5.2.4.7	3.4.9.7	x	x	x
Overload Protection	4.5.2.4.8	3.4.9.8	x		
Undervoltage Protection	4.5.2.4.9	3.4.9.9	x	x	
Voltage Output	4.5.2.4.10	3.4.9.10	x	x	x
Internal Noise	4.5.2.4.11	3.4.9.11	x	x	x
Input Current	4.5.2.4.12	3.4.9.12	x		x
Conducted RF	4.5.2.4.13	3.4.9.13.1	x		x
Magnetic Susceptibility	4.5.2.4.14	3.4.9.14	x		x
Ripple Susceptibility	4.5.2.4.15	3.4.9.15	x		x
RF Susceptibility	4.5.2.4.16	3.3.5	x		
Radiated R.F.	4.5.2.4.17	3.4.9.13.2	x		x
Voltage Transients Susceptibility	4.5.2.4.18	3.4.9.16	x		x
Temperature Altitude	4.5.2.4.19		x		
Humidity	4.5.2.4.19		x		
Salt Spray	4.5.2.4.19		x		
Vibration	4.5.2.4.19		x		*
Shock	4.5.2.4.19		x		*
Fungus Resistance	4.5.2.4.19		x		
Life	4.6		x	**	
***Reliability Assurance	4.7				

NOTES: *These tests shall be performed on the sample interconnecting box from the first ten production units and on the sample units for each fifth lot. Amplifiers subjected to this test shall not be submitted for acceptance under the contract.

**Life tests shall be made as per schedule of 4.6.4 and not on Individual or Sampling Test basis.

***Reliability Assurance Tests shall be made per schedule of 4.7.1 and 4.7.2

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4.5 General Test Procedure

4.5.1 Test Equipments - All instruments used in performing the measurements required by this specification shall be maintained and calibrated to assure quality production and shall meet at least one of the following requirements.

4.5.1.1 Preferred Standard Equipment - They shall be listed as preferred standard equipment in NAVWEP 16-1-525.

4.5.1.2 Specific Approval - They shall be acceptable to the cognizant Government Inspector and the procuring activity as indicated by written approval.

4.5.1.3 Audio Oscillator - The audio oscillator shall have a frequency range of at least 50 to 100,000 cps, shall have a high degree of stability in both output voltage and frequency and shall have a waveform distortion of less that 0.5 percent.

4.5.1.4 Vacuum Tube Voltmeter - The vacuum tube voltmeters used shall have flat frequency response (±1db) from at least 50 to 100,000 cps and shall be capable of measuring voltages from 1 microvolt to 100 volts rms.

4.5.1.5 Distortion Analyzer - The distortion analyzer shall be of the wave analyzer or total distortion type. The frequency range shall be at least 100 to 10,000 cps with high selectivity.

4.5.2 Test Methods - The minimum test requirements of this specification shall be conducted in accordance with the following procedures:

4.5.2.1 Examination of Product - Each interconnecting box shall be examined to determine conformance with this specification with respect to material, workmanship, design, weight and dimensions.

4.5.2.2 Operational Test - Each interconnecting box shall be operated with the combinations of equipments and circuits for which designed. See Figure 3. The period of operation shall be long enough to permit the temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory operation. The frequency of the source shall be varied over the 300 to 7000 cps audio range.

4.5.2.3 Overload Measurement - Audio power of 500 milliwatts rms at 1000 cps shall be applied through each headset circuit shown in Figure 2 to the required dummy load for a period of 8 hours. During the last hour of each test, the circuit shall be checked for temperature rise or signs of failure. Upon completion of the overload test, circuits shall be rechecked by the operational test of 4.5.2.2.

4.5.2.4 Microphone Circuit Including Preamplifier Assembly -

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4.5.2.4.1 Voltage Gain - The voltage gain shall be determined at 1000 cps using the test circuit in Figure 4. The voltage gain shall conform to the limits specified in 3.4.9.1.

4.5.2.4.2 Gain Stabilization - The gain stabilization time shall be measured to ensure compliance with the requirements of 3.4.9.1.1.

4.5.2.4.3 Impedance - Input impedance shall be measured at 1000 cps and shall meet the requirements of 3.4.9.2. The output impedance shall be measured as follows:

(1) Remove 100 ohm resistor from the test circuit in Figure 4 and insert an inductor having an impedance greater than 1000 ohms and a DC resistance equal to or less than 100 ohms; a series resistance shall be added to obtain a total DC resistance of 100 ohms. Couple the output to a resistance decade with a capacitor having a minimum capacitance of 60 microfarads.

(2) Set resistance decade to infinity and measure output voltage of .4 volt rms (reference).

(3) Set resistance decade to obtain one-half output voltage (.2 volt rms).

(4) The resistance value indicated by the decade at this point is the output impedance of the preamplifier and shall be as described in 3.4.9.2.

4.5.2.4.4 Linearity - The voltage gain of the preamplifier shall be measured with the input voltages at 10, 200 and 500 microvolts. Linearity measurements shall be in accordance with 3.4.9.4.

4.5.2.4.5 Frequency Response - Response measurements shall be made using the test circuit in Figure 4. Measurements shall be made at the frequencies specified in Table 1 and the response measurements shall conform to 3.4.9.5.

4.5.2.4.6 Distortion - The distortion shall be measured at 300 cps, 1000 cps and 2500 cps using the test circuit in Figure 4. Distortion measurements shall conform to 3.4.9.6.

4.5.2.4.7 Amplifier Stability - The preamplifier shall be tested for conformance with the stability requirements of 3.4.9.7.

4.5.2.4.8 Overload Protection - The preamplifier shall meet the gain specification of 3.4.9.1 upon removal of the overload condition specified in 3.4.9.8. The preamplifier shall be capable of withstanding the overload condition for fifteen minutes.

4.5.2.4.9 Under Voltage Protection - The preamplifier shall be tested for conformance with the requirements of 3.4.9.9.

4.5.2.4.10 Voltage Output - The minimum voltage output level shall be measured for conformance with 3.4.9.10 using the test circuit shown in Figure 4.

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4.5.2.4.11 Internal Noise - Noise shall be measured at the output terminals of the preamplifier under test with a voltmeter having a bandwidth greater than 100 kc. The noise voltage measured shall then be referenced to the input and shall conform to requirements of 3.4.9.11 when tested using the test circuit of Figure 4.

4.5.2.4.12 Input Current - Input current shall be measured at rated output power for conformance to 3.4.9.12.

4.5.2.4.13 Conducted RF - The preamplifier output shall be measured to ensure compliance with the requirements of 3.4.9.13.1, Table II. The test circuit shown in Figure 4 shall be used.

4.5.2.4.14 Magnetic Susceptibility - The preamplifier shall be tested for conformance with the requirements of 3.4.9.14.

4.5.2.4.15 Ripple Susceptibility - The preamplifier shall be tested to ensure conformance to the requirements of 3.4.9.15.

4.5.2.4.16 Radio Frequency Susceptibility - These tests shall be performed in accordance with Specification MIL-STD-461, 462, and 463 except that the frequency range shall extend from 0.15 to 1000 mc.

4.5.2.4.17 Radiated R.F. - The preamplifier output shall be measured to ensure compliance with the requirement of 3.4.9.13.2 Table II A. The test circuit shown in Figure 4 shall be used.

4.5.2.4.18 Voltage Transients Test - The preamplifier shall be subjected, through its output connections, to 350 volt pulses having a width of approximately 500 microseconds and a repetition rate of 2 p.p.s. The pulses shall be applied for a period of five minutes in each polarity. The preamplifier shall be inactive during this test. See (3.4.9.16).

4.5.2.4.19 Environmental Test - Environmental tests listed in Table III shall be performed in accordance with Specification MIL-T-5422.

4.5.2.5 Headset Circuits Including Audio Transformers

4.5.2.5.1 Frequency Response - The frequency response of each 15 ohm headset circuit of the interconnecting box shall be measured as follows unless otherwise required by the specification applicable to the required headset. Tests shall be made by checking the interconnecting box from the cable connector pins. Distortion measurements of 4.5.2.5.2 shall be made concurrently with frequency response measurements.

(1) Connect the input leads of the interconnecting box to the appropriate output terminals of the audio oscillator which have a value equal to the input impedance of the earphone (normally 600 ohms). In the event the audio oscillator has no corresponding resistive load-rated terminal, an impedance matching device may be used, provided there is sufficient output to overcome the loss.

(2) Connect the proper resistor, of a value equal to the impedance of the earphone, across each pair or pairs of the earphone output terminals of the interconnecting box. Refer to the schematic diagram in Figure 2.

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(3) Adjust the output from the audio oscillator to give 1.9 volts rms at 1000 cps as measured by a VTVM at the required output terminals to the earphones.

(4) Maintain the audio oscillator output established for the 1000 cps reading and repeat the measurement for each of the following frequencies:

(a) From 300 cps to 1000 cps inclusive, measure at each increment of 100 cps.

(b) From 1250 cps to 7000 cps inclusive, measure at each increment of 250 cps.

4.5.2.5.2 Distortion - The distortion introduced in the headset circuits shall be determined as follows:

(1) Connect the headset circuit of the interconnecting box as outlined in 4.5.2.5.1.

(2) Connect the distortion analyzer input in parallel with the VTVM.

(3) Measure the distortion introduced by the transformer for an output of 1.9 volts rms at 1000 cps.

(4) Maintain the audio oscillator output established for the 1000 cps reading and repeat the distortion measurements for each of the following frequencies:

(a) From 300 cps to 1000 cps inclusive-measure at each increment of 200 cps.

(b) From 2000 cps to 7000 cps inclusive-measure at each increment of 1000 cps.

4.5.2.5.3 Impedance Measurement - The impedance measurements shall be made at frequencies selected as necessary to insure that the requirements of 3.4.10.4 are met. The voltmeter-ammeter method of measurement or other method satisfactory to the cognizant Government Inspector shall be used. The voltage applied across the terminals during impedance measurements shall be approximately the same as applied under maximum output conditions. When measuring the impedance of the primary windings, the secondary windings shall be terminated with 15 ohms ±10% non-inductive resistors and when measuring the impedance of the secondary windings, the primary windings shall be terminated with 300 ohms ± 1% non-inductive resistors.

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4.6 Life Tests - The interconnecting box shall be subjected to a 300 hour life test at the contractor's plant under standard conditions as described in 3.3.8.

4.6.1 Test Data - The contractor shall keep a daily record as to the performance of the interconnecting box, making particular note of any deficiencies or failures. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part Type Number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to the failure.

4.6.1.1 Failure Report - In the event of a failure the Government Inspector shall be notified and the cause of the failure immediately determined. If the cause of the failure indicates material or design corrections are necessary, the U.S. Naval Ammunition Depot shall determine if such corrective action is necessary prior to acceptance of additional interconnecting boxes.

4.6.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically the interconnecting box shall be turned on and off several times and put through all phases of its performance.

4.6.3 Performance Check - At approximately 24 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the U.S. Naval Ammunition Depot.

4.6.4 Life Test Schedule - The interconnecting box to be tested shall be representative of the current production in all significant aspects of design and manufacture. It shall be selected by the Government Inspector from interconnecting boxes which have passed the Individual Tests. The Life Test shall be performed on the following quantities unless otherwise stated in the contract or approved by the U.S. Naval Ammunition Depot.

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Quantity of Interconnecting Boxes Offered for Acceptance	Quantity to be Tested
First 25	1
Next 26 to 200	1
Next 201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.7 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted.

4.7.1 Qualification Phase - Prior to the acceptance of interconnecting boxes under the contract or order, a minimum of three (3) interconnecting boxes shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of interconnecting boxes to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan II shall be used.

4.7.2 Reliability Production Acceptance (Sampling) Phase Tests - Interconnecting boxes throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test level E shall be used.

4.7.2.1 Lot Size for Sampling Phase - Interconnecting Boxes constructed during 1 month shall be one lot. The Accept-Reject criteria for Test Plan IV shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract or order, no interconnecting boxes shall be shipped until an Accept decision is reached under each lot.

4.7.3 Test Data - The contractor shall keep a daily record of the performance of the interconnecting boxes, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

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4.7.3.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.8 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.9 Reconditioning of Test Equipment - Interconnecting boxes which have been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the interconnecting boxes for acceptance.

4.10 Presubmission Testing - No item, part or complete interconnecting box shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.11 Rejection and Retest - Interconnecting boxes which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the interconnecting box shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the interconnection boxes are not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appedix D of MIL-STD-794.

6. NOTES

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6.1 Intended Use - It is intended that this interconnecting box be used in Naval Aircraft to permit the intercommunication set and/or radio to be operated with an optional combination of headsets and microphones.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (See 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this interconnecting box. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompained by complete supporting information.

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6.5 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer to the Federal Government tested for Qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the U.S. Naval Ammunition Depot, Crane Indiana 47522.

6.6 Associated Equipment - The Interconnecting Box shall operate with the following accessories, which will not be supplied as part of this equipment:

Cord Assembly - Impedance Matching Network	CX-4620/AR
Cord Assembly	CX-4621/AR
Cord Assembly	CX-4622/AR
Cord Assembly	CX-4623/AR

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TABLE A-INPUT CONNECTOR-J1	
PIN	FUNCTION
A	AUDIO INPUT NO. 1
B	EARPHONE COMMON
C	AUDIO INPUT NO. 2 OR KEY
D	MICROPHONE
E	MICROPHONE COMMON
F	SPARE

TABLE B-OUTPUT CONNECTOR-J2	
PIN	FUNCTION
A	RIGHT EARPHONE
B	EARPHONE COMMON
C	LEFT EARPHONE OR KEY
D	MICROPHONE
E	MICROPHONE COMMON
F	SPARE

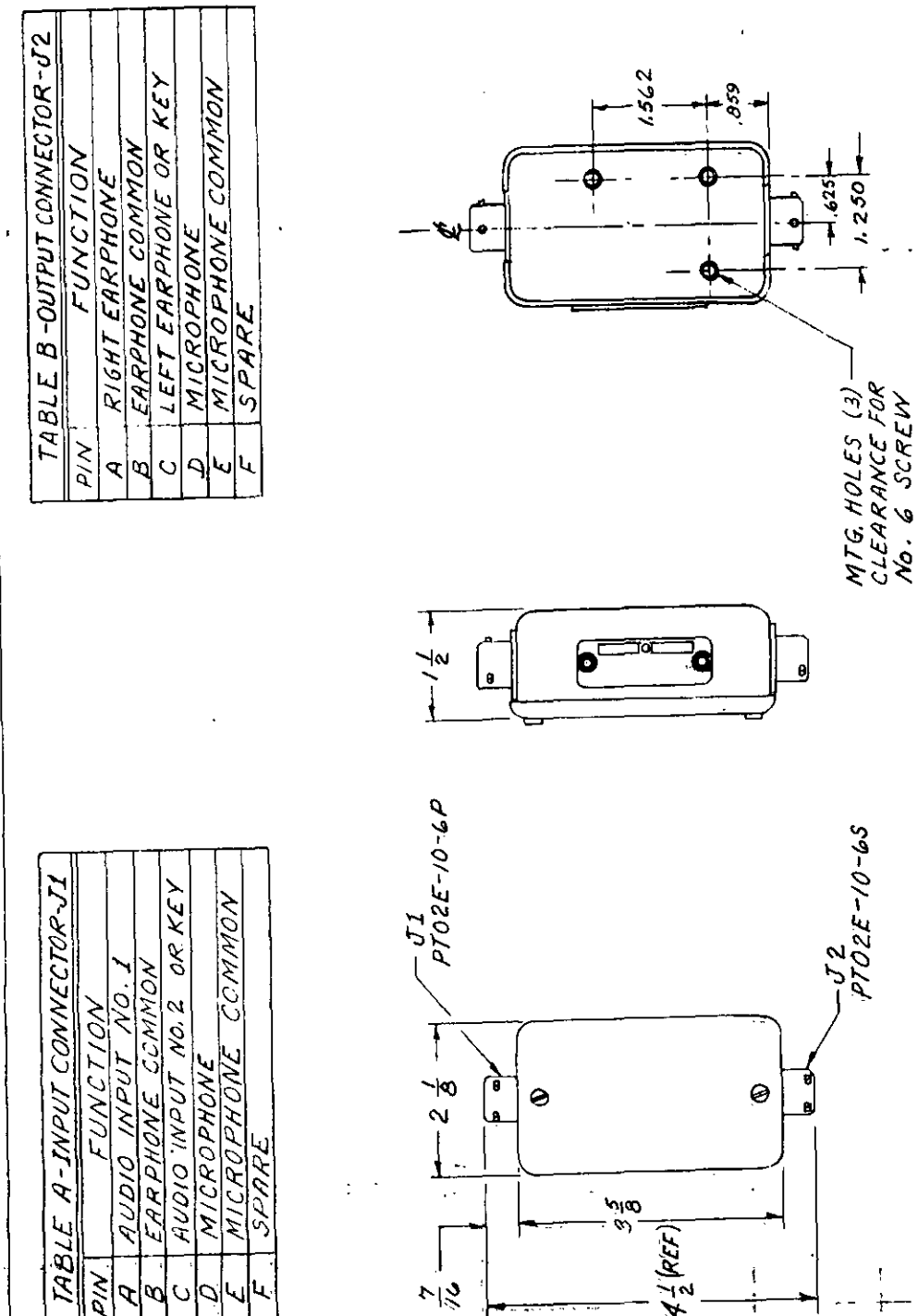
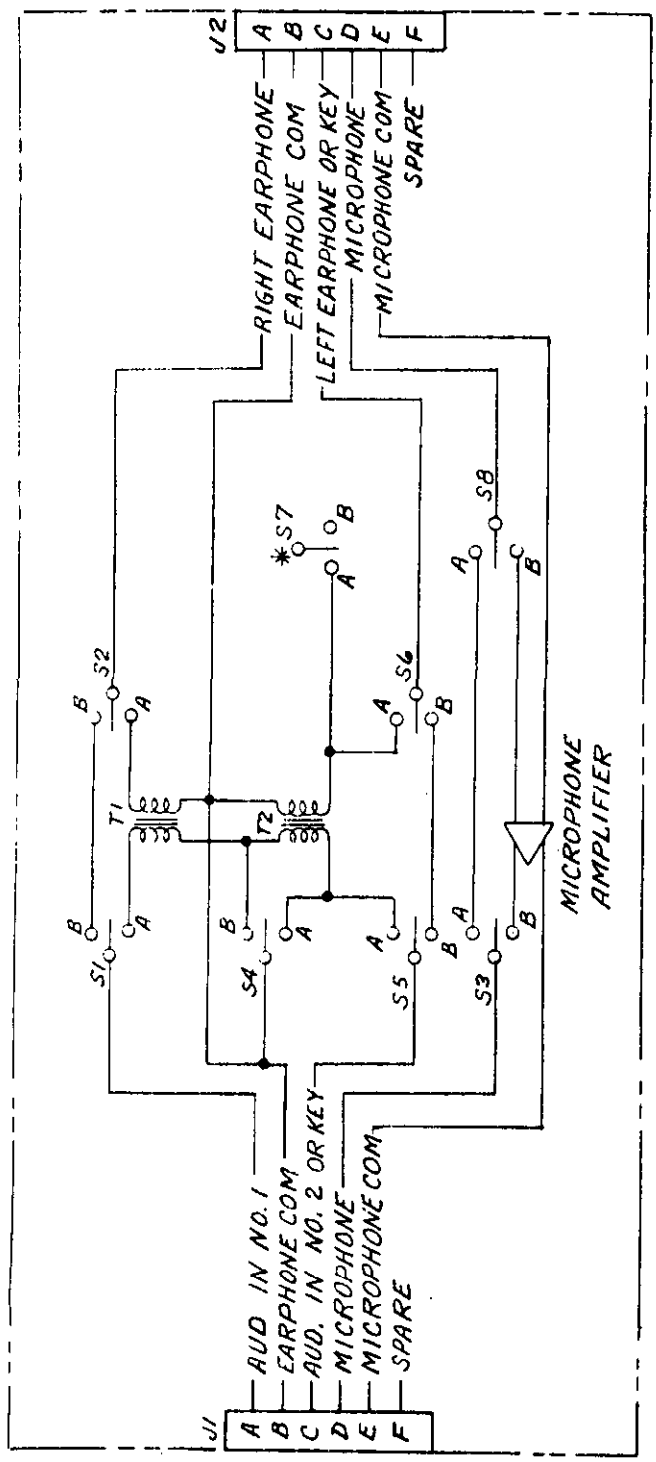


FIGURE 1. Outline Drawing of Interconnecting Box, J-1013A/AIC

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HEADSET COMBINATIONS	LINK NUMBER 1	2	3	4	5	6	7	8
IN-LINE KEYING SWITCH	LINK POSITIONS							
15 OHM PARALLEL	B	B	A	B	B	B	*	A
600 OHM PARALLEL	B	B	A	B	B	B	*	A
EXTERNAL KEYING SWITCH								
15 OHM PARALLEL	A	A	B	A	B	A	*	B
15 OHM SPLIT	A	A	B	B	A	A	*	B
600 OHM PARALLEL	B	B	A	B	B	B	*	A
600 OHM SPLIT	B	B	A	B	B	B	*	A

(SEE NOTES)

NOTES:
1. ALL SWITCHES ARE OF THE LINK TYPE.
2. SWITCH NUMBERS CORRESPOND WITH LINK NUMBERS SHOWN ON THE CHART.
* 3. SWITCH S7 NOT USED.

FIGURE 2. SCHEMATIC AND OPERATING CHART OF INTERCONNECTING BOX, J-1013A/AIC

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HEADSET AND MICROPHONE COMBINATIONS

HEADSET	MICROPHONE	KEYING LINE	MATCHING TERMINAL Z AT PINS OF CONNECTORS J1 PINS	J1 OHMS	J2 PINS	J2 OHMS
15 ohm parallel	5 ohm dynamic	See Note 1	A,B D,E	600 100	A,B B,C,15 D,E	5
600 ohm parallel	100 ohm carbon	See Note 1	A,B D,E	600 100	A,B D,E	600 100
15 ohm parallel	5 ohm dynamic	See Note 2	A,B D,E	600 100	A,B C,B D,E	15 15 5
15 ohm split	5 ohm dynamic	See Note 2	A,B C,B D,E	300 300 100	A,B C,B D,E	15 15 5
600 ohm parallel	100 ohm carbon	See Note 2	A,B D,E	600 100	A,B D,E	600 100
600 ohm split	100 ohm carbon	See Note 2	A,B C,B D,E	600 600 100	A,B C,B D,E	600 600 100

Note #1 - Pins C of connectors are available for a keying line with this operation.

Note #2 - Keying line is not available for use with this operation.

FIGURE 3: INPUT-OUTPUT CHART OF INTERCONNECTING BOX J-1013A/AIC

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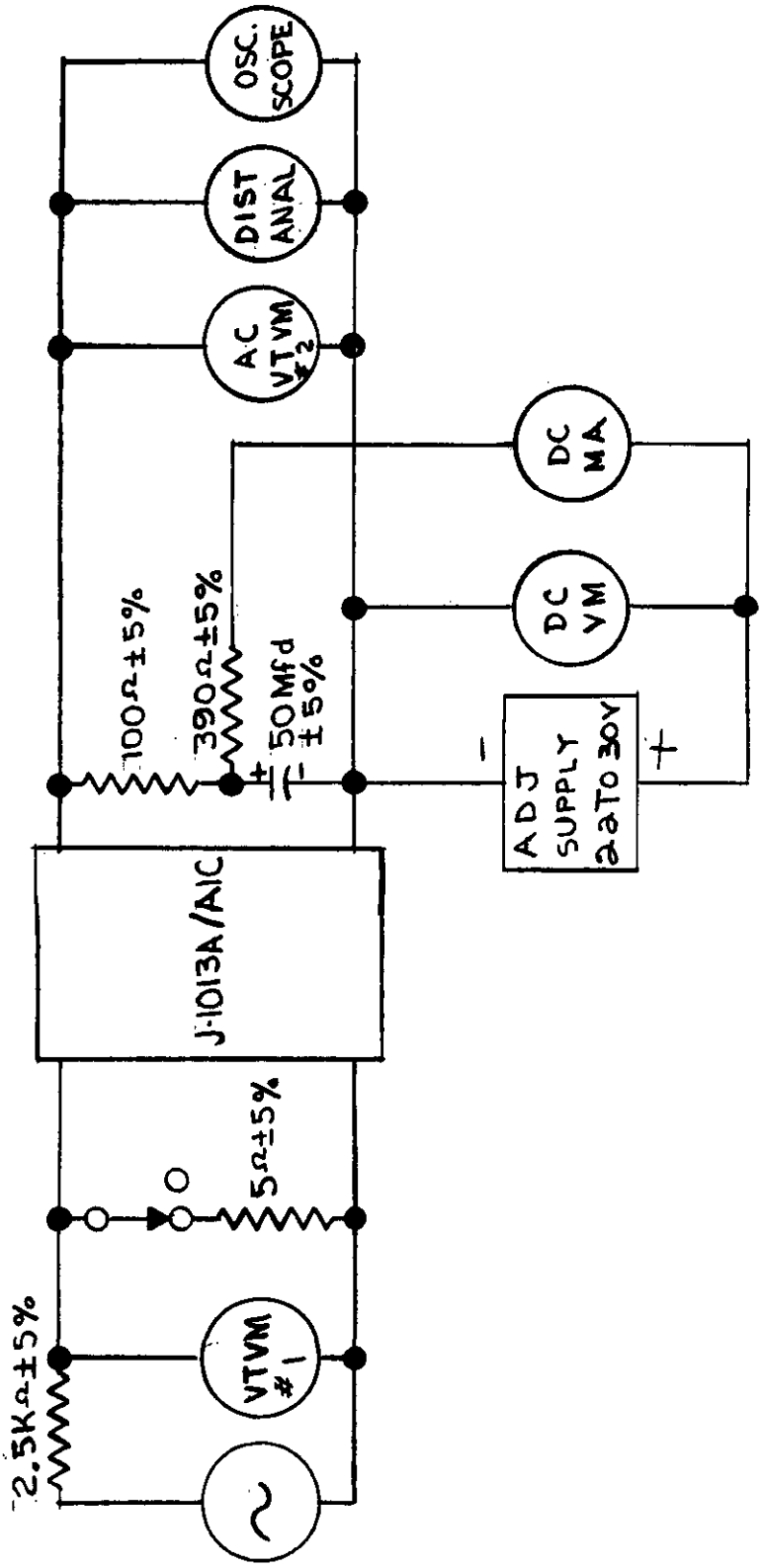


FIGURE 4
GENERAL TEST CIRCUIT OF PREAMPLIFIER OF
J-1013A/AIC

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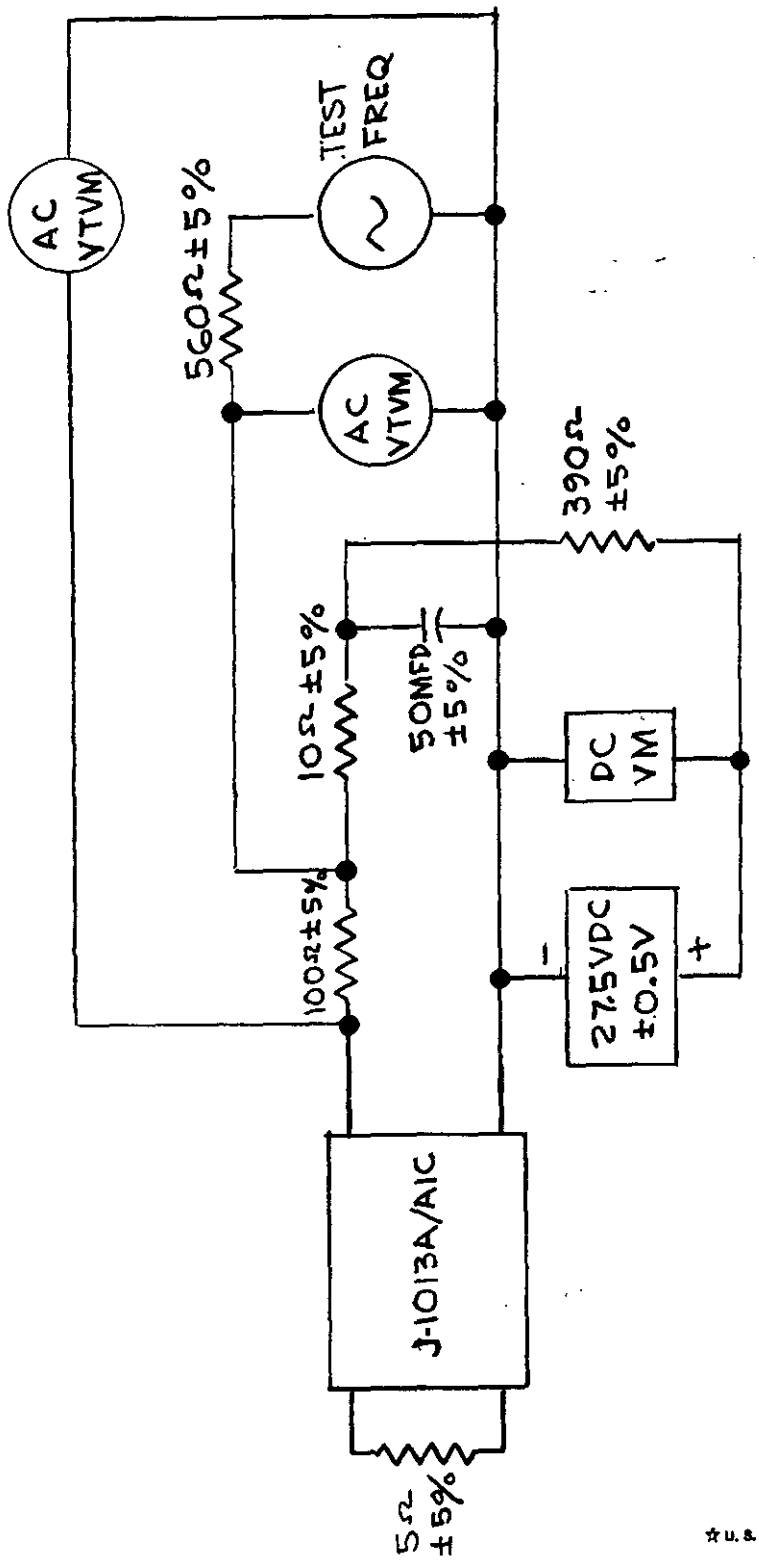


FIGURE 5
RIPPLE SUSCEPTIBILITY TEST CIRCUIT,
PREAMPLIFIER OF J-1013A/AIC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-I-22062A(AS) INTERCONNECTING BOX J-1013/AIC

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426

C-6870

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Naval Air Systems Command
Washington, D.C. 20360

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